RetailStream Data Platform

Project Execution Plan (PEP)

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1. Project Goal

Build an enterprise-grade batch data engineering platform that simulates a modern retail company's analytics ecosystem using open-source technologies.

The platform will:

- Generate realistic retail data
- Simulate operational systems
- Ingest batch data
- Process data using Medallion Architecture
- Build an analytical warehouse
- Create dashboards and reports
- Monitor infrastructure and pipelines
- Demonstrate production-grade engineering practices

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2. Development Methodology

The project is divided into **15 Epics**.

Each Epic contains multiple tasks.

Each task has:

- Objective
- Deliverable
- Dependencies
- Validation Criteria

No new feature will begin until the previous one is complete and tested.

Epic 1 – Foundation

Objective

Create the project's reusable foundation.

Tasks

FOUND-001

Create Git Repository

Deliverable

- GitHub Repository
- README
- License
- .gitignore

FOUND-002

Create Folder Structure

Deliverable

- Project folders
- Standard layout
- Empty modules

FOUND-003

Python Project Setup

Deliverable

- Virtual Environment
- requirements.txt
- pyproject.toml
- Common dependencies

FOUND-004

Configuration Framework

Deliverable

- YAML configuration
- Environment management
- Configuration loader

FOUND-005

Logging Framework

Deliverable

Reusable logging package.

FOUND-006

Utility Framework

Deliverable

Common helper modules.

FOUND-007

Documentation Structure

Deliverable

docs/ folder

Epic 2 – Data Generator Framework

Objective

Build a configurable framework capable of generating millions of realistic records.

Tasks

DATA-001

Generator Engine

DATA-002

Configuration Driven Generator

DATA-003

Indian Master Data

DATA-004

Customer Generator

100,000 rows

DATA-005

Category Generator

50 categories

DATA-006

Brand Generator

DATA-007

Supplier Generator

200 suppliers

DATA-008

Warehouse Generator

30 warehouses

DATA-009

Product Generator

50,000 products

DATA-010

Inventory Generator

300,000 records

DATA-011

Orders Generator

1,000,000 orders

DATA-012

Order Items Generator

4–5 million rows

DATA-013

Dirty Data Injection

DATA-014

Incremental Data Generator

Validation

- Referential integrity
- Configurable row counts
- Realistic business rules
- Reproducible output
- CSV, JSON, and Parquet support

Epic 3 – Infrastructure

Objective

Deploy the entire platform using Docker.

Tasks

INF-001

Docker Compose

INF-002

PostgreSQL

INF-003

MinIO

INF-004

Apache Spark

INF-005

Apache Airflow

INF-006

Apache Superset

INF-007

Prometheus

INF-008

Grafana

INF-009

dbt

INF-010

Health Checks

Validation

All services start successfully using one command.

Epic 4 – Source Systems

Objective

Create simulated production systems.

Tasks

SRC-001

FastAPI Framework

SRC-002

Customers API

SRC-003

Products API

SRC-004

Orders API

SRC-005

CSV Source Systems

SRC-006

Operational Database

Validation

APIs expose data correctly.

Epic 5 – Ingestion Framework

Objective

Build reusable ingestion pipelines.

Tasks

ING-001

Generic Ingestion Framework

ING-002

REST API Ingestion

ING-003

CSV Ingestion

ING-004

Metadata Capture

ING-005

Audit Logging

ING-006

Retry Logic

ING-007

Error Handling

Validation

Raw data lands successfully in Bronze.

Epic 6 – Airflow

Tasks

AIR-001

Master DAG

AIR-002

Customers DAG

AIR-003

Products DAG

AIR-004

Orders DAG

AIR-005

Inventory DAG

AIR-006

Supplier DAG

AIR-007

Bronze DAG

AIR-008

Silver DAG

AIR-009

Gold DAG

AIR-010

Warehouse DAG

AIR-011

Reporting DAG

Validation

Pipelines execute in dependency order.

Epic 7 – Bronze Layer

Tasks

- Raw ingestion
- Delta tables
- Metadata
- Partitioning
- Audit logs

Validation

Raw data matches source.

Epic 8 – Silver Layer

Tasks

- Deduplication
- Standardization
- Validation
- Data Quality
- Incremental processing

Validation

Business-clean data.

Epic 9 – Gold Layer

Tasks

- Sales KPIs
- Customer KPIs
- Product KPIs
- Inventory KPIs
- Supplier KPIs

Validation

Business-ready datasets.

Epic 10 – Warehouse

Tasks

- Star Schema
- Fact Tables
- Dimension Tables
- Indexes
- Constraints
- Warehouse Loader

Validation

Warehouse supports analytical queries.

Epic 11 – dbt

Tasks

- Staging Models
- Intermediate Models
- Mart Models
- Tests
- Documentation

Validation

All dbt tests pass.

Epic 12 – Reporting

Tasks

Executive Dashboard

Sales Dashboard

Customer Dashboard

Inventory Dashboard

Product Dashboard

Pipeline Dashboard

Validation

Business KPIs match warehouse data.

Epic 13 – Monitoring

Tasks

Infrastructure Metrics

Pipeline Metrics

Spark Metrics

Airflow Metrics

Warehouse Metrics

Alerts

Validation

Operational dashboards display system health.

Epic 14 – Testing

Tasks

Unit Tests

Integration Tests

Spark Tests

API Tests

Pipeline Tests

Data Quality Tests

Validation

All automated tests pass.

Epic 15 – CI/CD & Documentation

Tasks

GitHub Actions

Docker Build

Linting

Deployment Validation

README

Architecture Guide

Developer Guide

Setup Guide

Troubleshooting Guide

Demo Video

Validation

Repository is fully documented and deployable.

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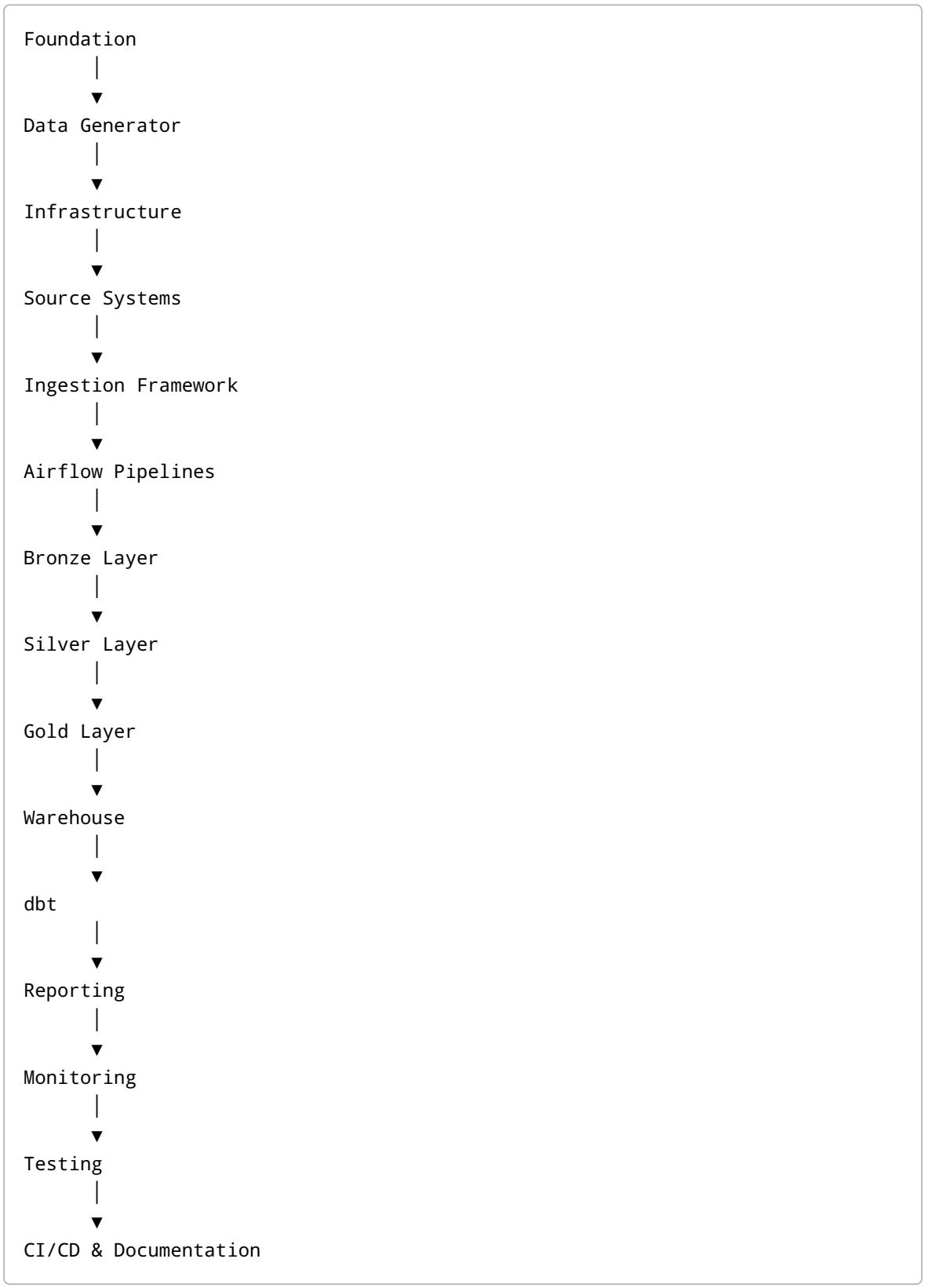
3. Project Dependency Flow



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4. Definition of Done (DoD)

A task is considered complete only if:

- Code is implemented.
- Unit tests pass.
- Integration tests pass (where applicable).
- Logging is implemented.
- Configuration is externalized.
- Documentation is updated.
- Code is committed using a meaningful Git commit.
- Code follows project standards.

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5. Final Deliverables

Upon completion, the project will include:

- Enterprise Data Generator
- Source System Simulation
- Batch Ingestion Framework
- Airflow Orchestration
- Medallion Data Lake
- Spark Processing Framework
- PostgreSQL Data Warehouse
- dbt Models
- Apache Superset Dashboards
- Prometheus & Grafana Monitoring
- Automated Testing
- CI/CD Pipeline
- Complete Documentation

The resulting repository will demonstrate the architecture, implementation, and operational practices expected in a modern enterprise data engineering platform.